

$$c_{new\ rms} = \sqrt{2}c_{rms}$$

15

D

From v - x graph, v is negative $\Rightarrow$ option C or D.

The gradient of the v - t graph corresponds to the acceleration.

Since $a = -\omega^2 x$, a and x are in opposite directions hence when x is negative, a is positive. Thus, the gradient of the v - t graph must be positive $\Rightarrow$ options A or D.

	Thus D meets both criteria.
16	C